Faizan Ahmed Syed

faizanahmedsyed1@gmail.com | LinkedIn: faizanahmed754 | github.com/TezorInnovate | techfaizan.com

Experience:

AUTONOMOUS DRONE SYSTEM | RESEARCH + ROBOTICS PROJECT | 2025-2026

- Desing and development of autonomous drone architecture.
- Complete assembly of F450 drone, esc's, flight controller configuration, and custom sensor array design.
- Raspberry-Pi based ROS2 node design for sensor integration and control via AI for automated flight patterns.

AI FIREFIGHTING ROVER | EMBEDDED SYSTEM DESIGN | JAN-FEB 2026

- Development of AI-based firefighting rover for detection and extinguishing process.
- Fine-tuned Yolov8 model for fire detection and custom software architecture pipeline.
- Custom designed hardware rover using esp32 based control, WiFi TCP protocol, and laptop-based AI compute.
- Testing, debugging, and troubleshooting complete hardware project for efficient functionality.
- Fully documented all project details in formal reports for referencing and explainability.

INTELLIGENT VISION-LANGUAGE ROBOT | ROBOTICS PROJECT | MAR 2026

- Built an AI system combining computer vision and LLM for real-time perception and response.
- Developed human-robot interaction features including natural language understanding and contextual response.
- Integrated sensors and control logic to enable autonomous environment awareness and adaptability.

TRAINEE ENGINEER | STRONG HAND CO. | JEDDAH, SAUDI ARABIA | MAY-JUNE 2023

- Contributed services to the Strong Hand Contracting Co. Jeddah Saudi Arabia as a trainee under the supervision of the consultant and lab specialist for the gas distribution system of the SOS Lab at Zahid Business Park.

Research Work and Publications:

AUTONOMOUS CYBER DEFENSE: AN AI-BASED SELF-HEALING FRAMEWORK (CCICT CONFERENCE PAPER)

- Random Forest based Cyber-Attack detection and mitigation using real time detection and healing actions.
- UNSW-NB15 IDS dataset used with data preprocessing and model training for accurate flagging.
- Modular software architecture design from detection to healing and response system.

BRAIN-COMPUTER INTERFACE ASSISTED AUTONOMOUS SYSTEMS USING DNN'S (UNDER REVIEW)

- Book chapter on BCI for autonomous systems and AI applications utilizing DNN's.
- Core software implementation of advanced BCI + DNN architecture for study and analysis.
- Detailed analysis of existing technology stacks with future scope and advancement possibilities.

DIABETES RISK PREDICTION USING ENSEMBLE LEARNING (UNDER REVIEW)

- In-depth study of existing ensemble models for diabetes risk prediction along with existing work.
- Model fine-tuning on available dataset after preprocessing and selection of best implementable architecture.
- Detailed results analysis with recent existing works and improvements in implementation processes.

Skills & Abilities:

- AI Stack: AIML, DL, CNN, GenAI, LLM
- Applied Research & Technical Problem Solving
- Programming – C/C++, Java, Python
- Microcontrollers: ESP32, Arduino, Raspberry Pi
- English C1 (IELTS - 7.5) Arabic B1 (High School Grade A+)
- Autonomous Systems & Robotics Integration
- AutoCAD, Autodesk Inventor, SolidWorks
- ROS2 Sensor Fusion and Gazebo Simulation
- Quick Learner, Hands-on assembly, Debugging
- Project planning and Technical Documentation

Education:

AMITY UNIVERSITY – NOIDA, INDIA (2023 – 2027 ON-GOING)

Bachelor of Technology: Computer Science and Engineering [ASET]
Current CGPA: 8.63 (cumulative GPA up to 5th semester)
Certifications: Software Testing, Python for Data Science, Advanced Robotics, Neural Networks for computer vision and NLP, AutoCAD Mastery.